

Hedieh Sazvar

PhD Candidate, Operations Research | Northwestern University
Optimization · Data Science · Experimentation

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Education

Northwestern University, Evanston, IL Sep 2021 – 2026 (Expected)
Ph.D. in Operations Research (GPA: 3.77)
Advisors: Prof. Seyed M.R. Iravani, Prof. Sanjay Mehrotra
M.S. in Industrial Engineering (GPA: 3.7), 2022
Sharif University of Technology, Tehran, Iran Sep 2016 – Feb 2021
B.S. in Industrial Engineering (GPA: 3.87)
Thesis: *IoT and Cloud Computing for Smart Agriculture* (Advisor: Prof. Mahmoud Houshmand)

Research Interests

Sequential decision-making and optimization under uncertainty, with applications to online matching platforms, recommender systems, and digital advertising. Methods span stochastic and integer optimization, dynamic programming, reinforcement learning, and large-scale experimentation.

Industry Experience

Moloco – Data Science Intern, Ads Data Science Jun 2025 – Sep 2025
Redwood City, CA

- Analyzed an A/B test on a core bidding component in Moloco's ad-serving stack, working with billions of auction-level events in BigQuery.
- Ran segment-level analysis across geography, platform, creative format, and campaign — found that flat aggregate results masked significant impact on top-spend campaigns.
- Developed a hypothesis-driven framework to diagnose drivers of underperformance: combined bid-level binning, win-rate decomposition, and predicted-vs-realized revenue analysis to isolate the mechanism behind the campaign-level effect.
- Identified a root cause in the interaction between the system's value-estimation logic and dynamic pricing signals; proposed a redesign projected to deliver double-digit ROAS gains for high-spend campaigns, along with an experimental design recommendation to improve baseline statistical power.
- Built the statistical inference layer using bootstrap resampling for confidence intervals across many segments.
- Engineered intermediate BigQuery tables to make terabyte-scale joins across bid, impression, and conversion data tractable; delivered a reproducible analysis pipeline the team continues to use.

Research Experience

Northwestern University – Graduate Research Assistant (NSF-funded) Jun 2022 – Present
Static and Dynamic Allocation for Online Matching Platforms

- Develop sequential decision and optimization models for two-sided online matching platforms (healthcare staffing, gig work) under demand uncertainty, supply no-shows, dependent preferences, and capacity constraints.
- Formulated mixed-integer and stochastic optimization models for sequential matching and personalized assortment under capacity constraints; designed scalable reformulations and solved high-degree binary polynomial problems via exact and heuristic methods (Python, Gurobi).
- Designed overbooking strategies for sequential allocation under no-show uncertainty, balancing fill rate against over-allocation risk.

Independent Research

Rollout-Augmented Lookahead Policies for Fork-Join Queues Spring 2025

- Developed lookahead policies augmented with simulation-based rollout for sequential information-gathering decisions in continuous-time queueing systems with partial observations; framed as a CTMC/MDP and benchmarked against threshold heuristics and tabular Q-learning.

Publications and Working Papers

- **H. Sazvar**, S. Iravani, S. Mehrotra. “Optimal Sequential Job Recommendations with No-Shows.” *In preparation for submission to Manufacturing & Service Operations Management*.
Formulates sequential job recommendation under no-shows as a 0–1 polynomial program; characterizes a threshold-optimal policy with a strongly polynomial-time exact algorithm for single-agent jobs and a nested-structure heuristic for multi-agent jobs. Quantifies the value of arrival-sequence predictability.
- **H. Sazvar**, S. Iravani, S. Mehrotra. “Overbooking in Sequential Job Recommendation Platforms with No-Shows.” *In preparation*.
Extends the framework to allow overbooking; develops a polynomial-time evaluation of the high-degree polynomial objective and embeds it in a MIP for exact solutions, with a structure-exploiting heuristic for large instances.
- **H. Sazvar**, S. Iravani, S. Mehrotra. “Optimization Models and Reformulations for Job Recommender Systems with Dependent Preferences and No-Shows.” *In preparation for submission to INFORMS Journal on Computing*.
Extends to dependent preferences via discrete choice models (multinomial logit, Markov chain choice, exponential); derives exact MIP reformulations and strengthens them with constructive heuristics and valid inequalities, demonstrating tractability at real-world scale.
- **H. Sazvar**. “Rollout-Augmented Lookahead Policies for Dynamic Join Decisions in Fork-Join Queues.” *In preparation*.
Studies when to stop collecting parallel-server signals before making a classification decision (delay vs. misclassification tradeoff); develops a k -step lookahead policy with H -step simulation-based rollout, benchmarked against threshold heuristics and tabular Q-learning.

Conference Presentations

- INFORMS Annual Meeting, Atlanta, GA Oct 2025
- INFORMS Annual Meeting, Seattle, WA Oct 2024

Skills

Programming & Tools: Python, SQL, R, BigQuery, Git, Gurobi, AMPL, LaTeX

Optimization & Decision-Making: Linear, Integer, Nonlinear, Convex, and Stochastic Optimization; Dynamic Programming; Markov Decision Processes; Reinforcement Learning; Multi-Armed Bandits; Policy Rollout and Lookahead; Distributed Optimization

Statistics & Data Science: A/B Testing, Experimentation, Statistical Inference, Bootstrap, Causal Models, Time Series, Machine Learning, Bayesian Methods, Monte Carlo Simulation

Domain: Online Marketplaces, Digital Advertising, Revenue Management, Matching and Assortment, Recommender Systems

Teaching Experience

Northwestern University – Teaching Assistant

Sep 2022 – Present

- **MLDS 490-23 Healthcare Analytics (M.S.):** delivered lectures on forecasting, optimization, scheduling, survival analysis, and R-based implementations.
- **IEMS 304 Statistical Learning for Data Analysis (M.S.), IEMS 315 Stochastic Models (M.S.), IEMS 303 Statistics (M.S.), IEMS 201 Data-Driven Decision Making (M.S.):** graded, held office hours, provided feedback.

Selected Graduate Coursework

Optimization & OR: Linear Programming, Nonlinear Programming, Integer Programming, Convex Optimization, Distributed Optimization, Stochastic Optimization, Dynamic Programming, Logistics, Stochastic Processes, Stochastic Simulation

Machine Learning & Statistics: Reinforcement Learning, Deep Learning, Applied Mathematical Statistics, Predictive Analytics

Honors and Awards

- Academic Excellence Award, Industrial Engineering Department, Sharif University of Technology 2016 – 2020
- Ranked 348th in Iran’s National University Entrance Exam among 500,000+ participants 2016